



Utility Supply and Construction

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CVR · THE HYDAKER-WHEATLAKE CO. · POWER LINE SUPPLY

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Owner: CVR Engineering

INTEGRATED CAPITAL DELIVERY PROCESS

Engineering · Estimating · Procurement · Project Management | CVR & HWC

Applicability: Mandatory for projects $\geq$ \$500,000

Exception authority: SVP Engineering or COO

Process owner: CVR

Purpose: Executive summary — for CEO/COO approval

1 EXECUTIVE SUMMARY

CVR proposes one integrated process for how CVR and HWC take capital projects from design through construction closeout — engineering, estimating, procurement, and project management operating on a single cost and schedule backbone, run with the same **Capital Delivery** discipline our investor-owned utility clients apply to their own capital programs.

Today those functions sit in different organizations with manual handoffs between them. The cost of those seams is real: estimates built without design quantities, purchase orders placed without engineering verification, GP surprises discovered after the money is spent, and schedules that don't drive procurement. As our IOU clients (DTE and others) move toward open-book commercial arrangements, they are effectively auditing our project controls — predictable delivery is becoming a condition of winning the work, not a differentiator.

Decisions requested of the CEO / COO:

1. **Adopt the integrated process** as mandatory for all projects $\geq$ \$500,000, with exceptions approved by the SVP Engineering or COO.
2. **Transition the estimating team from HWC into CVR**, reporting to the SVP of Engineering.

3. **Stand up the dedicated CVR project management team** — PMP/PE-credentialed PMs, separate from the engineering leads, operating in the Capital Delivery model with P6.
4. **Endorse the engineer-verified CVR → PLS transmittal system** as the single channel for material procurement.
5. **Pilot on the New Baltimore program** ahead of the spring 2027 construction start, then roll out across CVR/HWC.

2 KEY DEFINITIONS

Term	Meaning in this document
Capital Delivery	An operating mindset borrowed from how investor-owned utilities run their capital programs: every project is managed as a financial investment, not just a construction job . In practice that means a baselined budget and schedule for every project, visibility of committed dollars (not just invoiced dollars), a monthly forecast-at-completion, and gated leadership decisions. For CVR/HWC, it means managing gross profit the way an IOU's capital program office manages rate-base spend — proactively, with standing reports instead of after-the-fact reconciliation.
IOU	Investor-Owned Utility — a shareholder-owned, commission-regulated utility (e.g., DTE Energy). IOUs fund construction through capital programs whose cost and schedule performance flows into their rate cases, which is why they expect open-book transparency and professional project controls from their partners.
EOR	Engineer of Record — the licensed engineer holding overall responsibility for the project design. In this process, the EOR is also the technical authority for all procurement content (part numbers, quantities, specifications).
GP	Gross Profit — contract revenue minus direct project costs (labor, material, equipment, subcontracts). The primary financial performance measure at the project and work-order level.
Committed cost	Dollars obligated by a purchase order or subcontract, whether or not invoiced yet. Tracking commitments — not just paid actuals — is what makes early GP forecasting possible.
Open book	A commercial arrangement in which the contractor shares its full cost buildup with the client — direct costs, overhead, markup/fee, and contingency — rather than a lump-sum price.
P6	Oracle Primavera P6 — the industry-standard scheduling platform for utility capital programs. Supports logic-driven critical-path schedules that carry resources and costs, not just dates.

Resource / cost loading	Assigning crews, staff, and dollars to schedule activities, so the schedule forecasts manpower needs and cash flow — and material need dates — rather than only showing durations.
Transmittal	A numbered, revision-controlled release of engineer-verified material requirements from CVR to PLS. The auditable record of what was ordered, when, and on whose authority.
Need date / lead time	The date material must be on site (from the P6 schedule) and the supplier's time from PO to delivery. PO placement deadline = need date minus lead time — the arithmetic that "buy on time, every time" runs on.
WBS / cost codes	A common work-and-cost breakdown structure shared by estimating, procurement, scheduling, and accounting, so estimate, POs, schedule, and actuals reconcile automatically instead of by hand.
RACI	Responsible / Accountable / Consulted / Informed — a role-clarity matrix. Exactly one Accountable owner per activity; Responsible parties do the work.

3 PURPOSE & BIG PICTURE

A **CVR-led process**, working closely with **HWC Construction/Operations Project Managers**, that:

1. **Builds accurate estimates** — the estimating team sits inside CVR, reporting to the SVP of Engineering (transitioning from HWC), building estimates from real design input and validating them with HWC Construction/Operations PMs who know field conditions and production rates.
2. **Leads procurement through Power Line Supply (PLS)** — equipment procurement is led by the **Lead Engineer, the Engineer of Record (EOR)** for the project, ensuring accurate part numbers, quantities, and technical requirements before anything is ordered.
3. **Runs projects with a dedicated CVR Project Manager** — a **PMP and/or PE, separate from the engineering lead** — operating in a **Capital Delivery** mindset: solid financial performance, on-time purchasing and ordering of equipment for the most efficient delivery possible, and professional tracking using P6 resource-loaded schedules.

The critical relationship: the interaction between the **CVR PM** and the **HWC Construction PM** is the backbone of this process. The CVR PM owns capital delivery — cost, schedule, procurement, reporting. The HWC Construction PM owns field execution — crews, production, constructability — and is **engaged from design kickoff, not mobilization**: constructability and field-operations input shapes the design and the estimate, not just the

build. Neither succeeds without the other: this pairing meets weekly, shares one schedule and one cost picture, and jointly owns the project outcome.

4 GOVERNANCE

Rule	Detail
Threshold	Projects $\geq$ \$500,000 must follow this process.
Exception authority	The SVP Engineering or the COO may approve an alternate approach for a given project.
Below threshold	Lightweight path applies; teams may still opt in to the full process.
Gate decision timing	At project award / kickoff, documented in the kickoff record.

5 ORGANIZATION & KEY ROLES

Role	Definition & Responsibility
Lead Engineer (EOR)	The Engineer of Record for the project. Owns design integrity and all technical procurement content — part numbers, quantities, technical requirements.
CVR Project Manager	Dedicated PMP/PE , a separate person from the EOR. Accountable for capital delivery end to end: financials, procurement timing, P6 schedule, reporting.
CVR Estimating	Part of CVR, reporting to the SVP of Engineering (org transition from HWC). Builds estimates on a shared cost-code structure.
HWC Construction PM	Field execution owner and constructability voice, involved from design kickoff through closeout — reviewing design for constructability and field operations before it hardens into an estimate.
PLS (Power Line Supply)	Procurement execution: sourcing, supplier management, PO placement per CVR-verified transmittals.
CVR PM Support Team	P6 scheduling, cost tracking, transmittal register, and reporting horsepower behind the CVR PM.
SVP Engineering / COO	Gate authority: process applicability, estimate sign-off, quarterly portfolio governance.

6 PROCESS FLOW

1 • Design — CVR Engineering (EOR) + HWC Construction/Ops input

Quantity takeoffs and technical specifications, with constructability and field-operations review from day one



2 • Estimate — CVR Estimating + HWC Construction PM review

Shared cost-code structure; constructability validated by the field



3 • Procurement — EOR-verified content → CVR-PLS Transmittal → PLS

PO timing driven by P6 schedule need dates



4 • Execution — CVR PM ↔ HWC Construction PM partnership

P6 resource-loaded schedule; cost tracked against the estimate



5 • Closeout

Lessons and actuals feed back into the estimating library

7 CVR ↔ PLS TRANSMITTAL SYSTEM

A controlled, numbered transmittal is the **only** channel by which material requirements move from CVR to PLS — no verbal orders, no email-thread BOMs.

Element	Content
Transmittal No.	[Project] -PLS-### sequential, with revision letter (e.g., NB-PLS-004 Rev B)
Bill of Material	Line-item part numbers, descriptions, quantities, units
Technical requirements	Specs, standards, ratings, approved manufacturers / equals
Need dates	Required-on-site dates pulled from the P6 schedule
Deliver-to	Site / laydown / warehouse instructions

Authorization	EOR signature (technical accuracy) + CVR PM signature (budget & schedule alignment)
PLS acknowledgment	Confirmation with pricing, lead times, and promise dates within 5 business days

Change control: any change to part number, quantity, spec, or need date requires a revised transmittal (new revision letter) with the same dual signature — a clean audit trail of what was ordered, when, and on whose authority.

Platform: standardized transmittal form + status register in SharePoint/Teams (Issued → Acknowledged → PO Placed → Delivered); graduate to a procurement module if volume justifies. The discipline matters more than the tool.

8 SYSTEMS ROADMAP

Current state: the process today runs on email and network drives. That cannot support committed-cost visibility, schedule-driven procurement, or auditable transmittals at portfolio scale — but the system decision must not stall the process launch. The recommendation: **process first, platform second** — start on tools we already own, run a structured two-vendor evaluation in parallel, and let the New Baltimore pilot generate the real requirements list.

PHASED APPROACH

Phase	What	Timing	ROM Cost*	Purpose
Crawl	SharePoint/Teams: transmittal form + register, WO cost workbook, standard templates	Now — 60 days	\$0 software (existing M365); ~\$10–15k internal effort	Prove the discipline immediately; generate the requirements list for the platform evaluation
Walk	Platform pilot on New Baltimore (Unifier vs. InEight evaluation below) + estimating tool selection	Q4 2026 – Q2 2027	\$75–150k first year (pilot licenses + partner support)	Stress-test against the 39-WO cost reconciliation before spring 2027 construction

Run	Full rollout: platform + P6 + estimating tool integrated; ERP/accounting feed for actuals	2027–2028	\$150–400k/yr run-rate depending on platform and seats	Estimate → PO → schedule → actuals on one backbone, GP forecast as a standing report
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PLATFORM OPTIONS

Option	Role in this process	P6 integration	ROM cost*	Fit assessment
Oracle Primavera Unifier	Capital program cost management: cost sheets tied to P6 WBS/CBS, cash flow, change management, configurable workflows (incl. the CVR-PLS transmittal)	Native, bi-directional	\$75–250k/yr subscription + \$100–300k implementation (4–9 mo, partner-led)	Strongest match to the Capital Delivery model ; the same class of tooling our IOU clients run. Heaviest lift.
InEight	Contractor-side suite; estimating-rooted; estimate → cost codes → project controls	Yes (connector)	\$50–150k/yr + \$50–150k implementation (3–6 mo)	The lighter-lift challenger — matches our estimating-led flow; evaluate head-to-head with Unifier.
Oracle Aconex	Controlled document transmittals & correspondence — heavyweight option for the CVR-PLS transmittal register and engineering doc control	Via Oracle ecosystem	\$30–100k/yr (project-based pricing)	Add-on consideration if transmittal volume outgrows SharePoint; not a standalone answer.
Procore	Field/construction PM, daily logs, RFIs; friendliest UX for HWC field teams	Connector (3rd-party)	\$40–120k/yr (priced on construction volume); light implementation (\$10–30k)	Strong at the HWC field end, shallow on capital-program cost controls. Possible complement, not the backbone.
Trimble e-Builder /	Owner-side capital program platforms	Varies	\$75k+/yr	Built for owners like DTE, not contractor

Kahua	operations — not recommended for CVR's backbone.
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ESTIMATING TOOLS (EVALUATION ALREADY UNDERWAY)

Tool	Strength	ROM cost*	Process fit
HCSS HeavyBid	Utility / heavy-civil estimating standard; crew-based production estimating	\$15–35k first year (small estimating team), ~\$10–20k/yr after	Exports structured estimates to seed P6 cost loading and the shared WBS
InEight Estimate	Deep integration with the InEight controls suite	\$20–50k/yr (team of 3–5)	Strongest if InEight wins the platform evaluation — one vendor, one data model
Excel (status quo)	Familiar; zero cost	\$0	No structured export, no version control, no shared WBS — the source of today's reconciliation pain

*ROM = rough order of magnitude, planning-level only — actual pricing varies with seat counts, modules, and construction volume, and several vendors price by quote only. Budgetary quotes from Oracle and InEight are a named deliverable of the Walk phase; no software commitment is requested in this document beyond the pilot evaluation.

DECISION PATH

1. Launch the Crawl phase on approval of this document (no new spend).
2. Fold the estimating tool evaluation already underway into the platform evaluation, so the estimate data model and the cost-controls data model are chosen together.
3. Bring budgetary quotes and the Unifier-vs-InEight recommendation to the quarterly Portfolio / Governance Review for a platform decision.

9 MEETING CADENCE

Meeting	Frequency	Required Attendees	Purpose
Project Kickoff / Process Gate	Once, at award	CVR PM, EOR, Estimating lead, HWC Construction	Confirm process applicability (≥\$500k), baseline

		PM, PLS rep (SVP Eng/COO if exception sought)	scope/budget/schedule, assign RACI names
CVR PM ↔ HWC Construction PM Sync	Weekly — never cancelled	CVR PM, HWC Construction PM (+ superintendent as needed)	3-week lookahead, crew/resource needs, field issues, schedule progress
Procurement & Expediting Review	Biweekly	EOR, CVR PM, PLS rep, PM Support scheduler	Transmittal register status, promise dates vs. need dates, expediting actions
Project Cost / GP Review	Monthly	CVR PM, Estimating lead, EOR, HWC Construction PM	Estimate vs. committed vs. actual by WO, GP forecast, trend/change log
P6 Schedule Update & Re-forecast	Monthly (feeds cost review)	PM Support scheduler, CVR PM, HWC Construction PM	Progress the schedule, re-forecast, update resource loading
Portfolio / Governance Review	Quarterly	SVP Engineering, COO, CVR PMs, Estimating lead	Portfolio GP performance, exceptions granted, lessons learned, process improvements

The weekly CVR PM ↔ HWC Construction PM sync is the anchor — everything else flexes around it. The monthly cost review uses the P6 update from the same week so cost and schedule tell one story.

10 RACI MATRIX

A Accountable **R** Responsible **C** Consulted **I** Informed | EOR = Lead Engineer (CVR) · EST = CVR Estimating
· PMS = CVR PM Support

Activity	EOR	CVR PM	EST	HWC PM	PLS	PMS	SVP	COO
Process applicability decision (≥\$500k / exception)	C	R	I	I	—	—	A	A
Design deliverables & quantity takeoffs	A /R	I	C	C	—	—	I	—
Estimate development	C	C	A /R	C	C	—	I	—
Constructability review of estimate	C	C	C	A /R	—	—	—	—

Estimate approval / bid sign-off	C	R	R	C	—	—	A	
Project baseline (budget + P6 schedule)	C	A/R	C	C		R		—
BOM: part numbers, quantities, tech requirements	A/R	C	C	C	C	—	—	—
Transmittal issuance to PLS	R	A/R				R	—	—
PO placement & supplier management	C		—		A/R		—	—
Expediting & promise-date management	C	A	—	C	R	R	—	—
Material receipt & field verification			—	A/R	C		—	—
P6 schedule updates & resource loading	C	A	—	C		R	—	—
Field execution (crews, production, safety)	C	C	—	A/R	—	—	—	
Cost tracking: estimate vs. committed vs. actual		A/R	C	C		R		
GP forecast & monthly project report		A/R	C	C	—	R		
Change management (scope / spec / quantity)	R	A/R	C	C				—
Client-facing commercial deliverables (e.g., open book)	C	A/R	R	C	—	R		
Closeout: actuals into estimating library, lessons learned	C	A	R	R		R		—
Process improvement & standards ownership	C	R	R	C	C	C	A	

- The **CVR PM is Accountable for capital delivery** end to end — the Capital Delivery mindset made structural.

- The **EOR is Accountable for everything technical** — the dual-signature transmittal is where the two accountabilities meet.
- The **HWC Construction PM is Accountable for the field** and Consulted on nearly everything upstream.
- **SVP / COO are Accountable only at the gates**, keeping leadership out of the day-to-day critical path.

11 SUCCESS CRITERIA

- Any work order's estimated vs. committed vs. actual cost is answerable without a manual reconciliation exercise.
- Material need dates in the P6 schedule drive PO placement dates automatically.
- Zero procurement rework from wrong part numbers or quantities — EOR verification and transmittal change control catch it before the PO.
- GP projection is a standing report, not a fire drill.
- Every $\geq \$500k$ project has a documented gate decision; exceptions are visible to the SVP/COO quarterly.



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